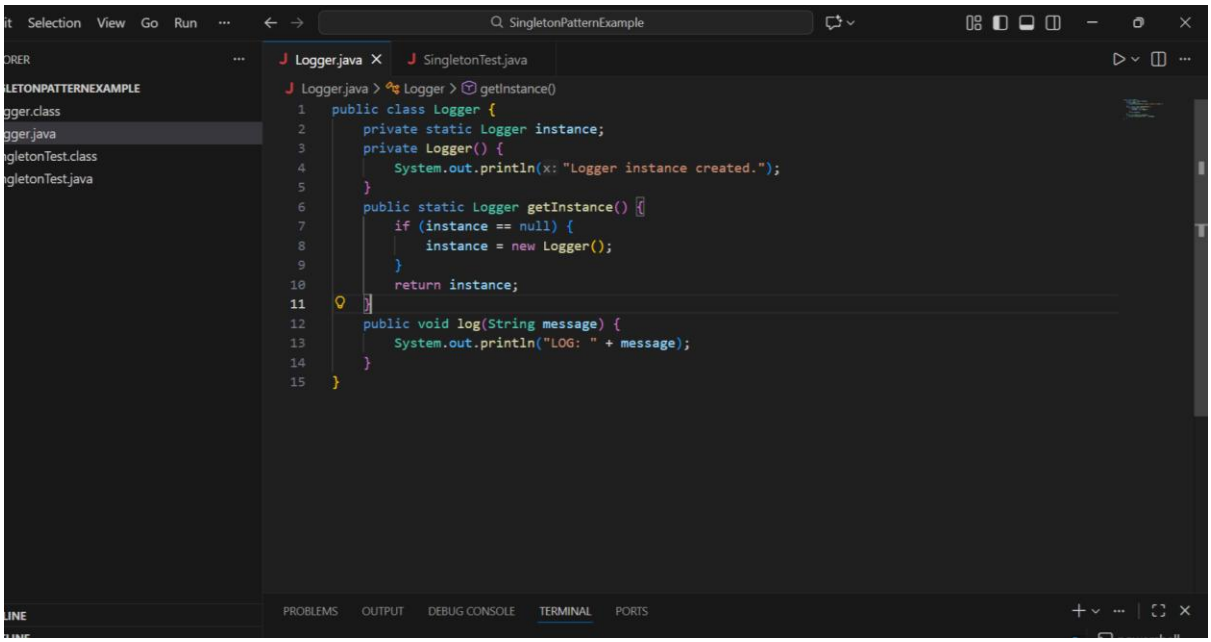
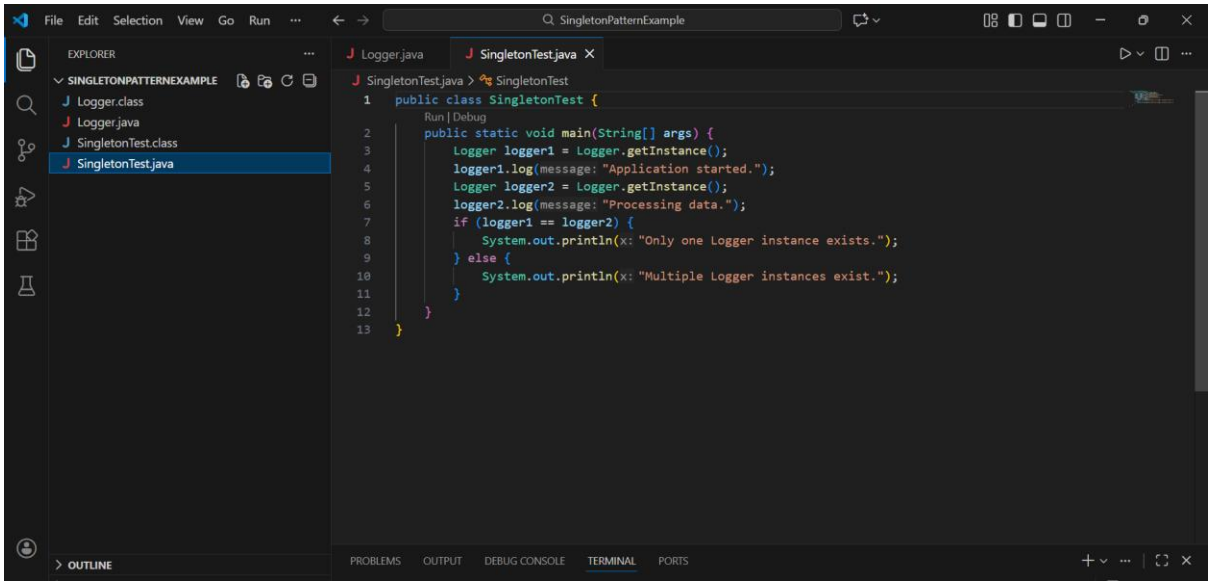
DIGITAL NURTURE 5.0 DEEPSKILLING

WEEK 1 – HANDS-ON

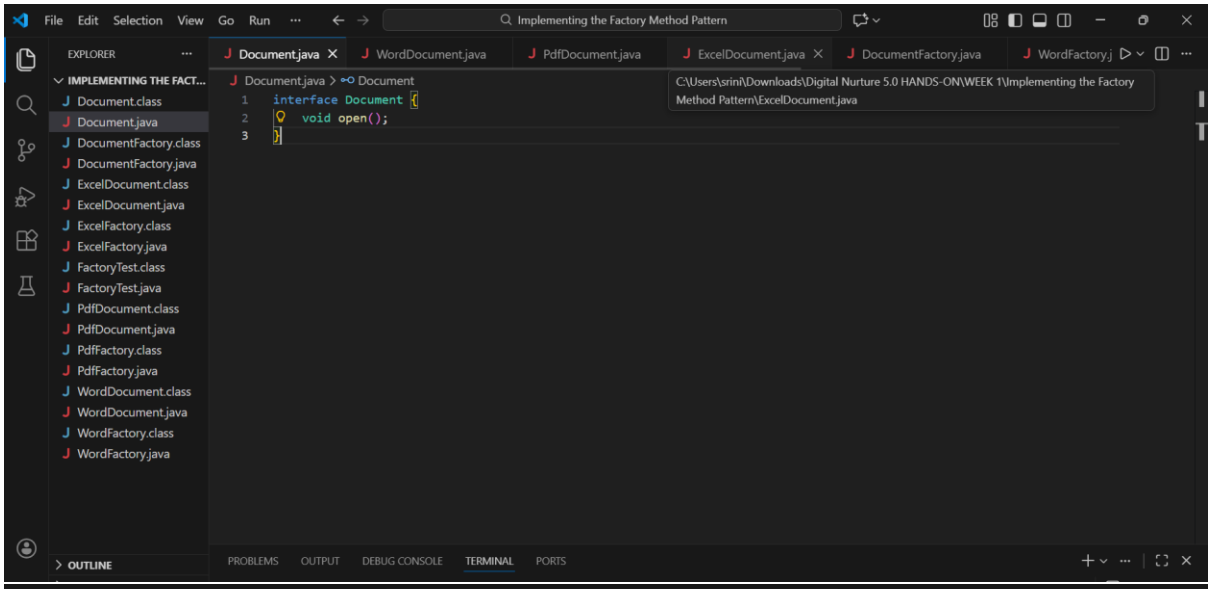
Design principles & Patterns

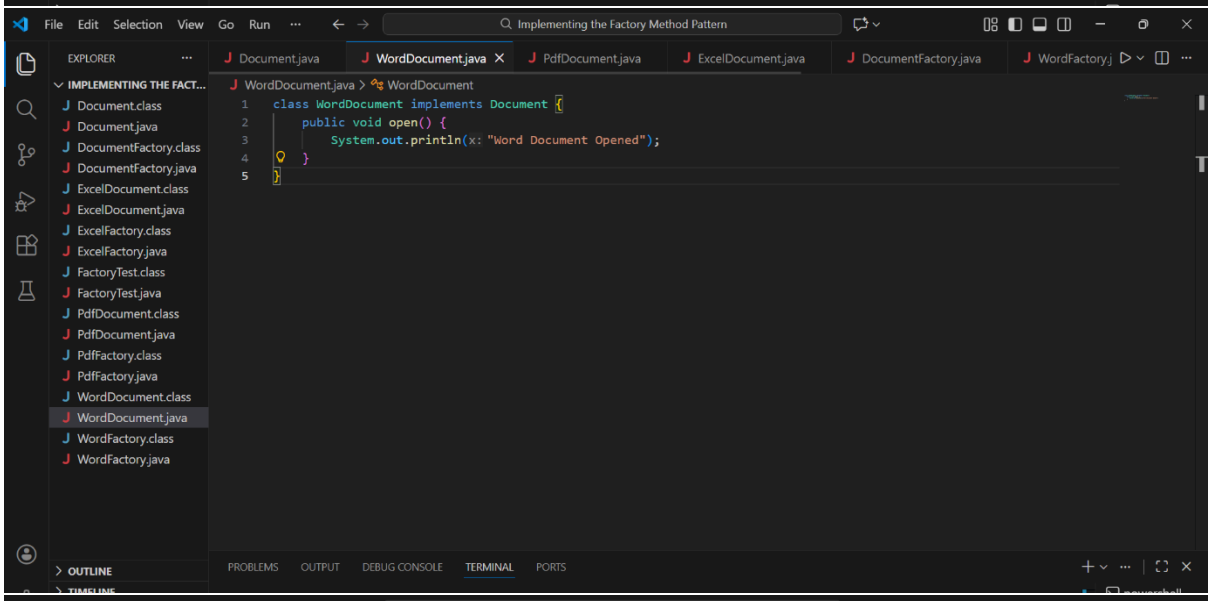
Exercise 1: Implementing the Singleton Pattern

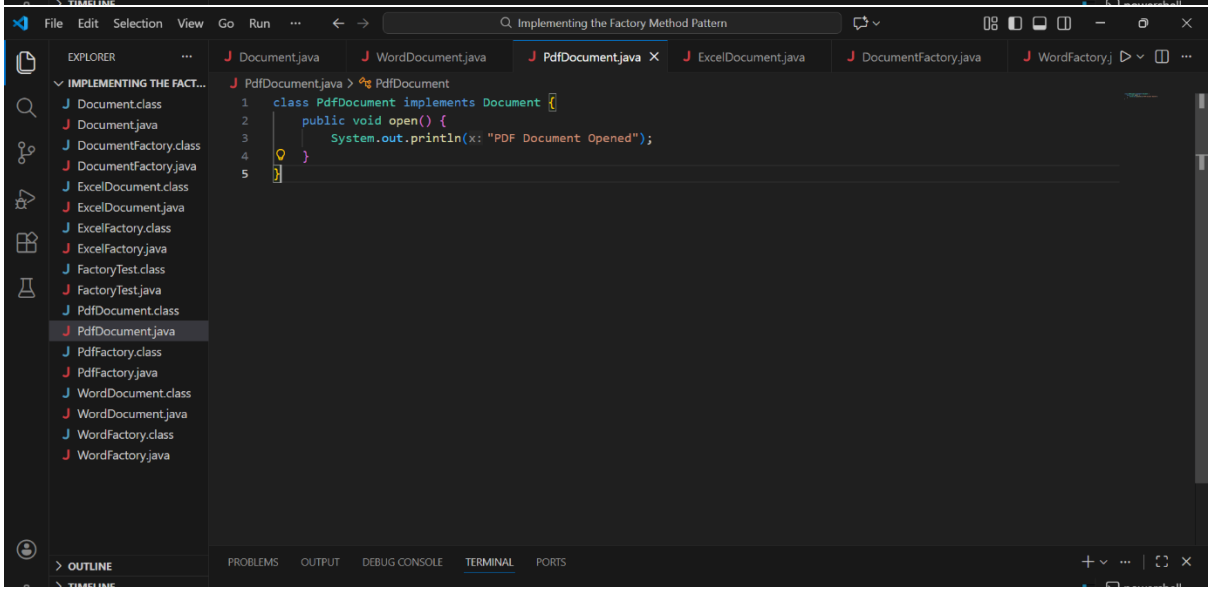




Exercise 2: Implementing the Factory Method Pattern


File Edit Selection View Go Run
Implementing the Factory Method Pattern
EXPLORER
IMPLEMENTING THE FACT...
Document.class
Document.java
DocumentFactory.class
DocumentFactory.java
ExcelDocument.class
ExcelDocument.java
ExcelFactory.class
ExcelFactory.java
FactoryTest.class
FactoryTest.java
PdfDocument.class
PdfDocument.java
PdfFactory.class
PdfFactory.java
WordDocument.class
WordDocument.java
WordFactory.class
WordFactory.java
Document.java
WordDocument.java
PdfDocument.java
ExcelDocument.java
DocumentFactory.java
WordFactory.j
Document.java > Document
C:\Users\srini\Downloads\Digital Nurture 5.0 HANDS-ON\WEEK 1\Implementing the Factory Method Pattern\ExcelDocument.java
1 interface Document {
2 void open();
3 }
OUTLINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS


File Edit Selection View Go Run
Implementing the Factory Method Pattern
EXPLORER
IMPLEMENTING THE FACT...
Document.class
Document.java
DocumentFactory.class
DocumentFactory.java
ExcelDocument.class
ExcelDocument.java
ExcelFactory.class
ExcelFactory.java
FactoryTest.class
FactoryTest.java
PdfDocument.class
PdfDocument.java
PdfFactory.class
PdfFactory.java
WordDocument.class
WordDocument.java
WordFactory.class
WordFactory.java
Document.java
WordDocument.java
PdfDocument.java
ExcelDocument.java
DocumentFactory.java
WordFactory.j
WordDocument.java > WordDocument
1 class WordDocument implements Document {
2 public void open() {
3 System.out.println(x: "Word Document Opened");
4 }
5 }
OUTLINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS


File Edit Selection View Go Run
Implementing the Factory Method Pattern
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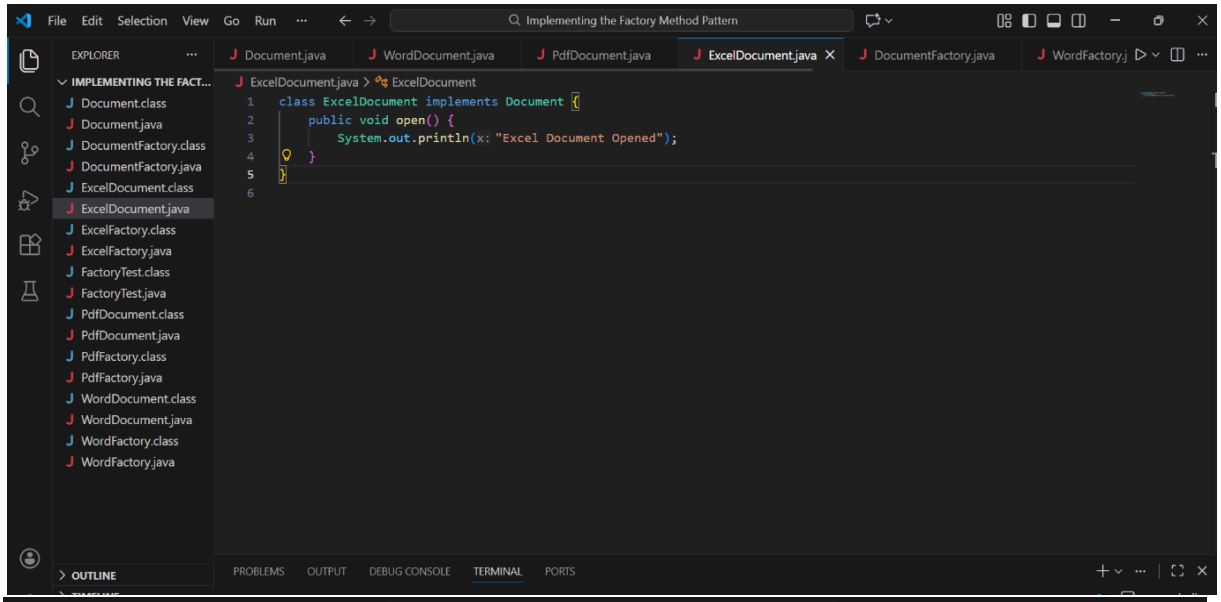
DocumentFactory.java
ExcelDocument.class
ExcelDocument.java
ExcelFactory.class
ExcelFactory.java
FactoryTest.class
FactoryTest.java
PdfDocument.class
PdfDocument.java
PdfFactory.class
PdfFactory.java
WordDocument.class
WordDocument.java
WordFactory.class
WordFactory.java
Document.java
WordDocument.java
PdfDocument.java
ExcelDocument.java
DocumentFactory.java
WordFactory.j
PdfDocument.java > PdfDocument
1 class PdfDocument implements Document {
2 public void open() {
3 System.out.println(x: "PDF Document Opened");
4 }
5 }
OUTLINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS


File Edit Selection View Go Run
Implementing the Factory Method Pattern
EXPLORER
IMPLEMENTING THE FACT...
Document.class
Document.java
DocumentFactory.class
DocumentFactory.java
ExcelDocument.class
ExcelDocument.java
ExcelFactory.class
ExcelFactory.java
FactoryTest.class
FactoryTest.java
PdfDocument.class
PdfDocument.java
PdfFactory.class
PdfFactory.java
WordDocument.class
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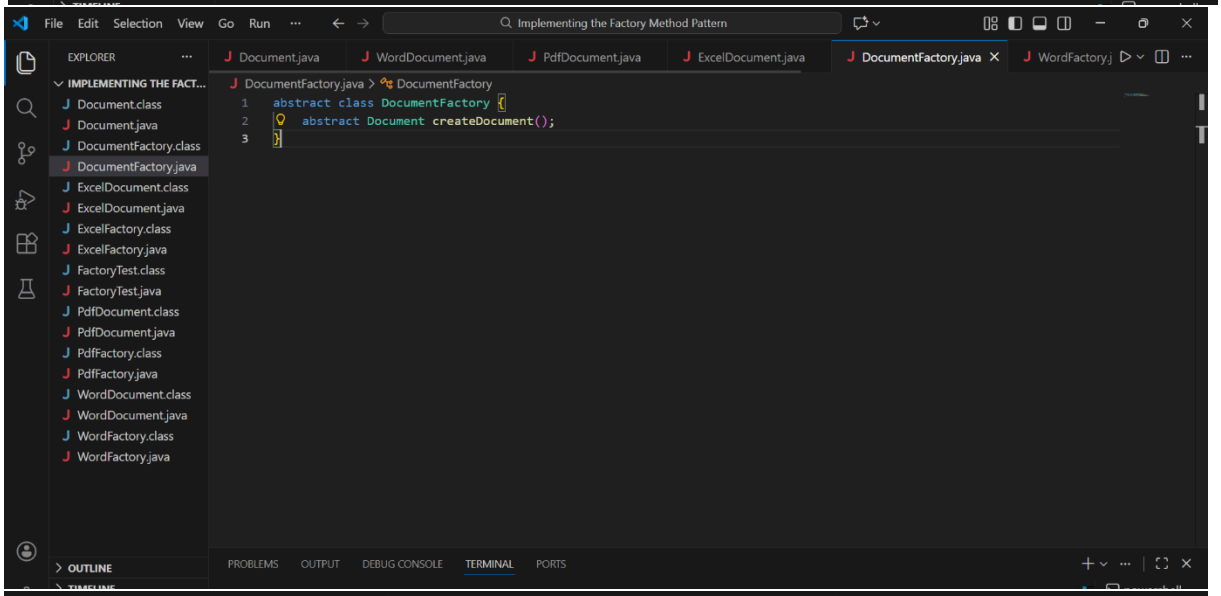
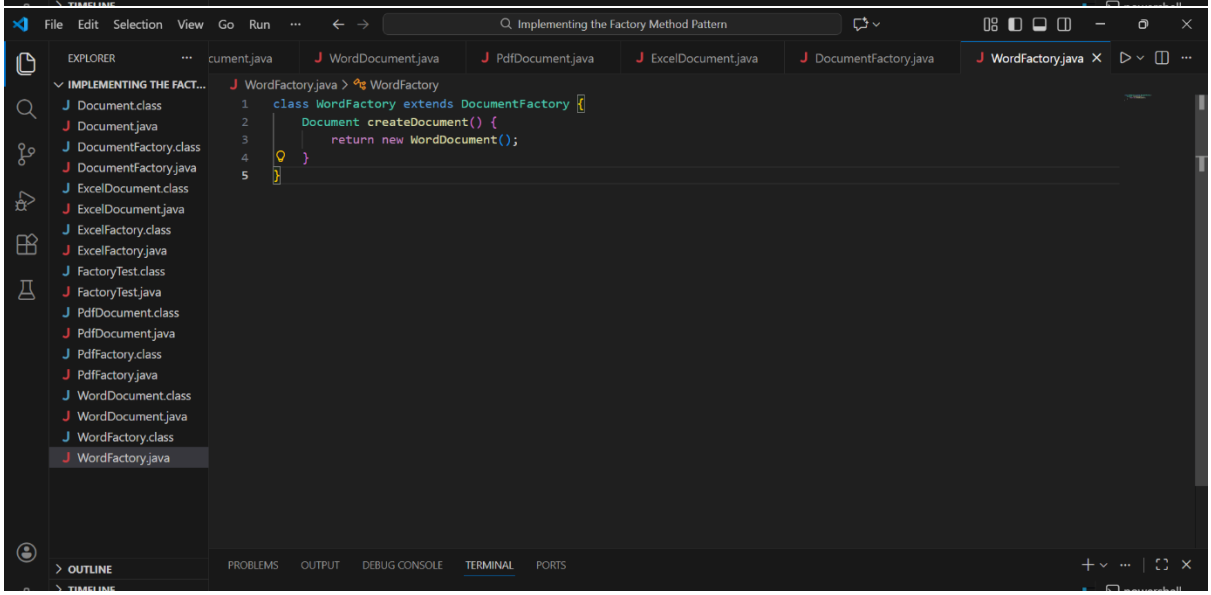
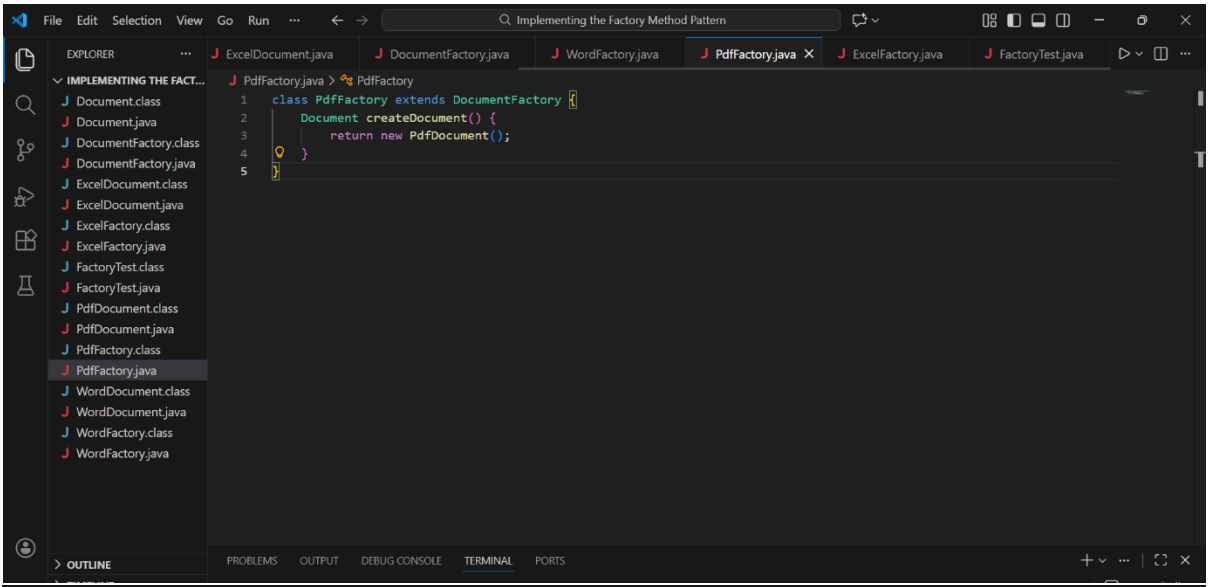
WordFactory.class
WordFactory.java
Document.java
WordDocument.java
PdfDocument.java
ExcelDocument.java
DocumentFactory.java
WordFactory.j
ExcelDocument.java > ExcelDocument
1 class ExcelDocument implements Document {
2 public void open() {
3 System.out.println(x: "Excel Document Opened");
4 }
5 }
6
OUTLINE
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS


File Edit Selection View Go Run
Implementing the Factory Method Pattern
EXPLORER
IMPLEMENTING THE FACT...
Document.class
Document.java
DocumentFactory.class
DocumentFactory.java
ExcelDocument.class
ExcelDocument.java
ExcelFactory.class
ExcelFactory.java
FactoryTest.class
FactoryTest.java
PdfDocument.class
PdfDocument.java
PdfFactory.class
PdfFactory.java
WordDocument.class
WordDocument.java
WordFactory.class
WordFactory.java
Document.java
WordDocument.java
PdfDocument.java
ExcelDocument.java
DocumentFactory.java
WordFactory.j
DocumentFactory.java > DocumentFactory
1 abstract class DocumentFactory {
2 abstract Document createDocument();
3 }
OUTLINE
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS


File Edit Selection View Go Run
Implementing the Factory Method Pattern
EXPLORER
IMPLEMENTING THE FACT...
Document.class
Document.java
DocumentFactory.class
DocumentFactory.java
ExcelDocument.class
ExcelDocument.java
ExcelFactory.class
ExcelFactory.java
FactoryTest.class
FactoryTest.java
PdfDocument.class
PdfDocument.java
PdfFactory.class
PdfFactory.java
WordDocument.class
WordDocument.java
WordFactory.class
WordFactory.java
cument.java
WordDocument.java
PdfDocument.java
ExcelDocument.java
DocumentFactory.java
WordFactory.java
WordFactory.java > WordFactory
1 class WordFactory extends DocumentFactory {
2 Document createDocument() {
3 return new WordDocument();
4 }
5 }
OUTLINE
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS


File Edit Selection View Go Run
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IMPLEMENTING THE FACT...
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Document.java
DocumentFactory.class
DocumentFactory.java
ExcelDocument.class
ExcelDocument.java
ExcelFactory.class
ExcelFactory.java
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FactoryTest.java
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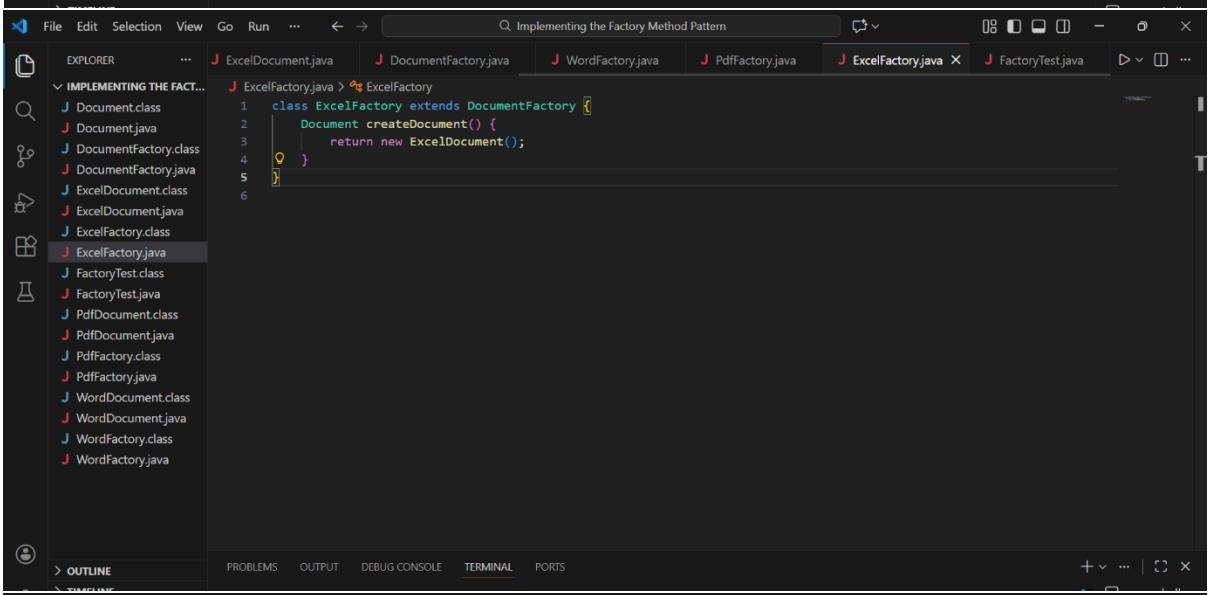
PdfDocument.java
PdfFactory.class
PdfFactory.java
WordDocument.class
WordDocument.java
WordFactory.class
WordFactory.java
ExcelDocument.java
DocumentFactory.java
WordFactory.java
PdfFactory.java
ExcelFactory.java
FactoryTest.java
PdfFactory.java > PdfFactory
1 class PdfFactory extends DocumentFactory {
2 Document createDocument() {
3 return new PdfDocument();
4 }
5 }
OUTLINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS


File Edit Selection View Go Run
Implementing the Factory Method Pattern
EXPLORER
IMPLEMENTING THE FACT...
Document.class
Document.java
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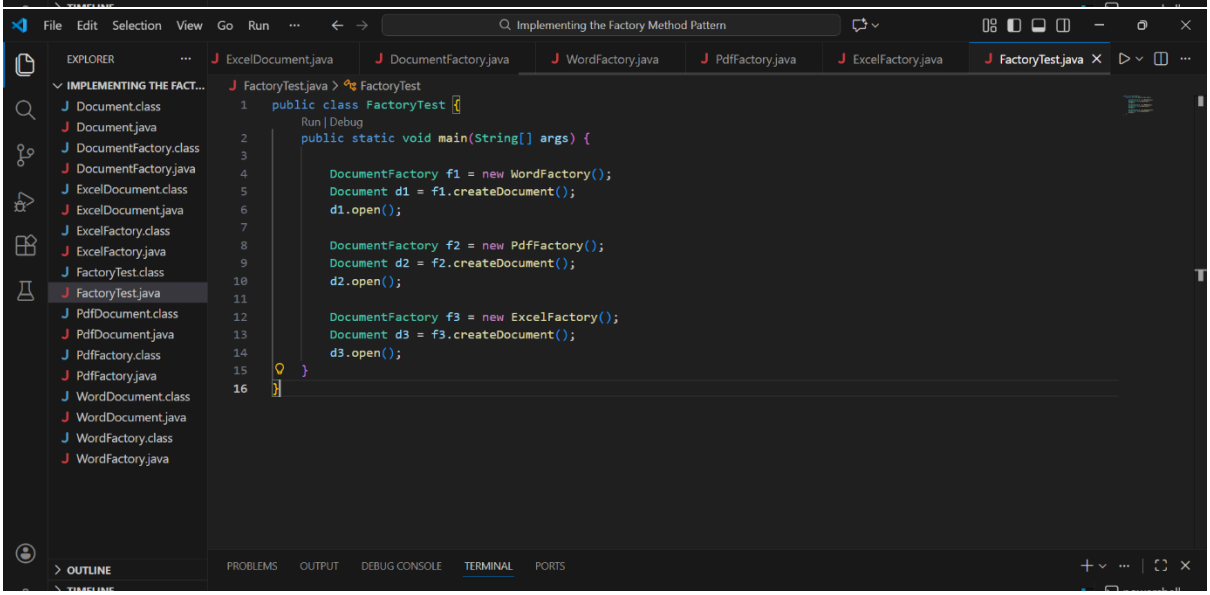
DocumentFactory.java
ExcelDocument.class
ExcelDocument.java
ExcelFactory.class
ExcelFactory.java
FactoryTest.class
FactoryTest.java
PdfDocument.class
PdfDocument.java
PdfFactory.class
PdfFactory.java
WordDocument.class
WordDocument.java
WordFactory.class
WordFactory.java
ExcelDocument.java
DocumentFactory.java
WordFactory.java
PdfFactory.java
ExcelFactory.java
FactoryTest.java
ExcelFactory.java > ExcelFactory
1 class ExcelFactory extends DocumentFactory {
2 Document createDocument() {
3 return new ExcelDocument();
4 }
5 }
6
OUTLINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS


File Edit Selection View Go Run
Implementing the Factory Method Pattern
EXPLORER
IMPLEMENTING THE FACT...
Document.class
Document.java
DocumentFactory.class
DocumentFactory.java
ExcelDocument.class
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FactoryTest.class
FactoryTest.java
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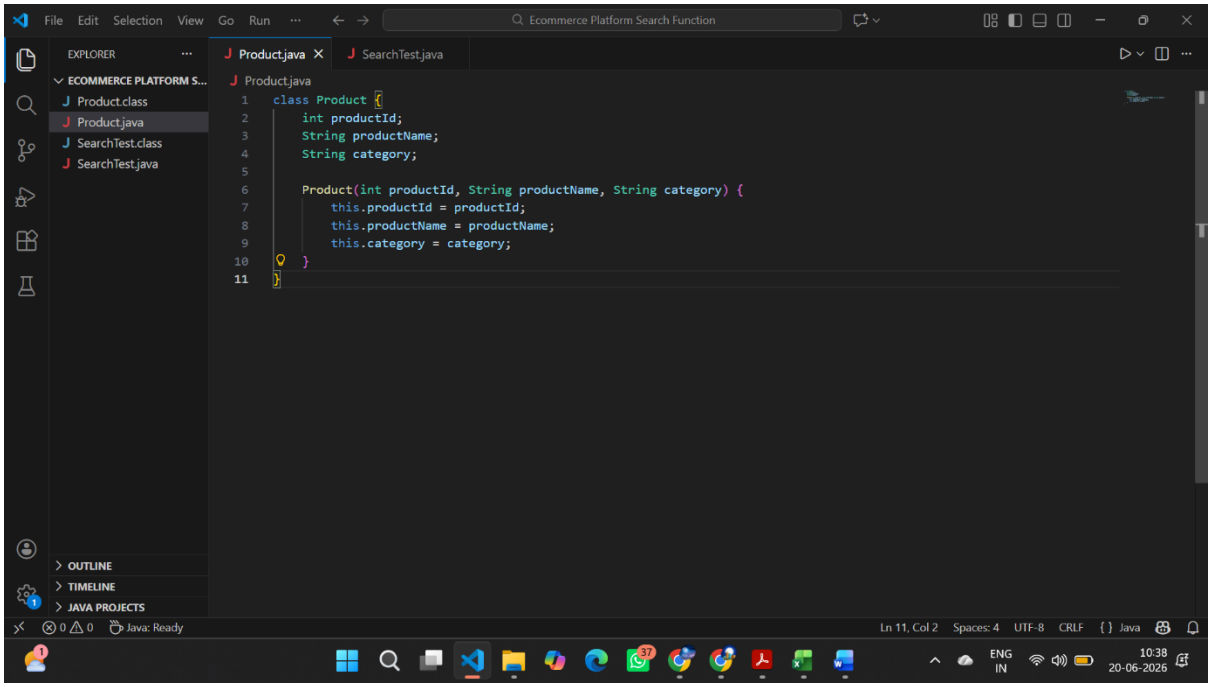
PdfDocument.java
PdfFactory.class
PdfFactory.java
WordDocument.class
WordDocument.java
WordFactory.class
WordFactory.java
ExcelDocument.java
DocumentFactory.java
WordFactory.java
PdfFactory.java
ExcelFactory.java
FactoryTest.java
FactoryTest.java > FactoryTest
1 public class FactoryTest {
Run | Debug
2 public static void main(String[] args) {
3
4 DocumentFactory f1 = new WordFactory();
5 Document d1 = f1.createDocument();
6 d1.open();
7
8 DocumentFactory f2 = new PdfFactory();
9 Document d2 = f2.createDocument();
10 d2.open();
11
12 DocumentFactory f3 = new ExcelFactory();
13 Document d3 = f3.createDocument();
14 d3.open();
15 }
16 }
OUTLINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Data structures and Algorithms

Exercise 2: E-commerce Platform Search F



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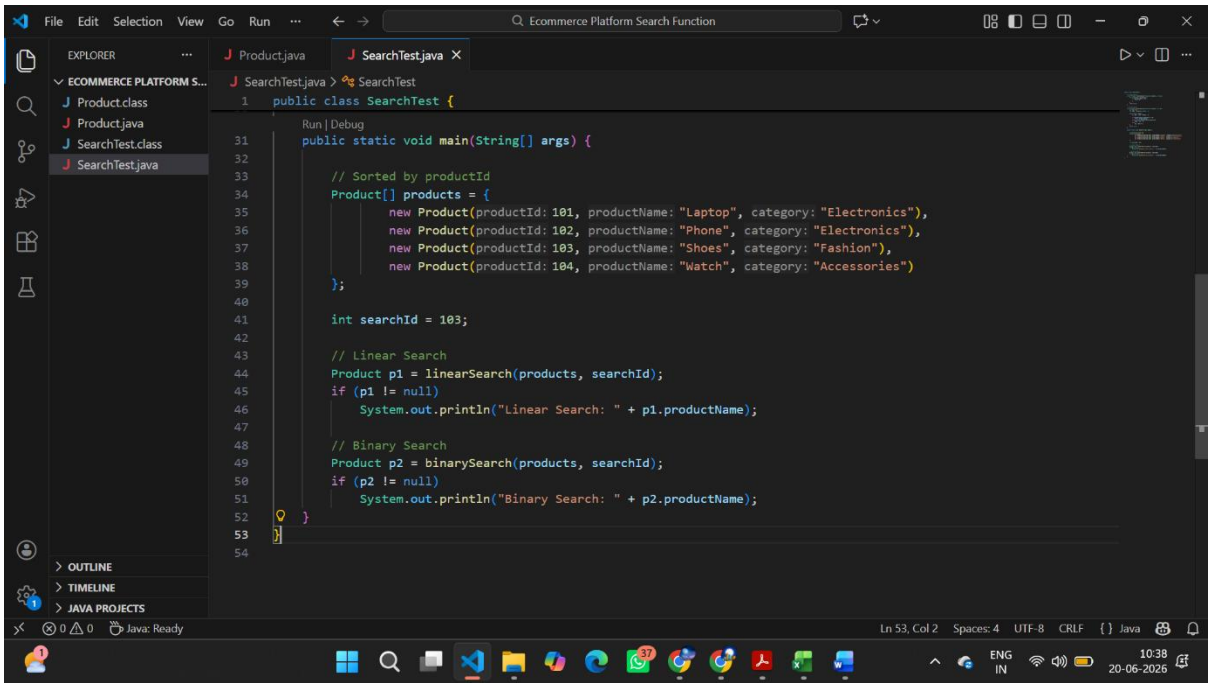
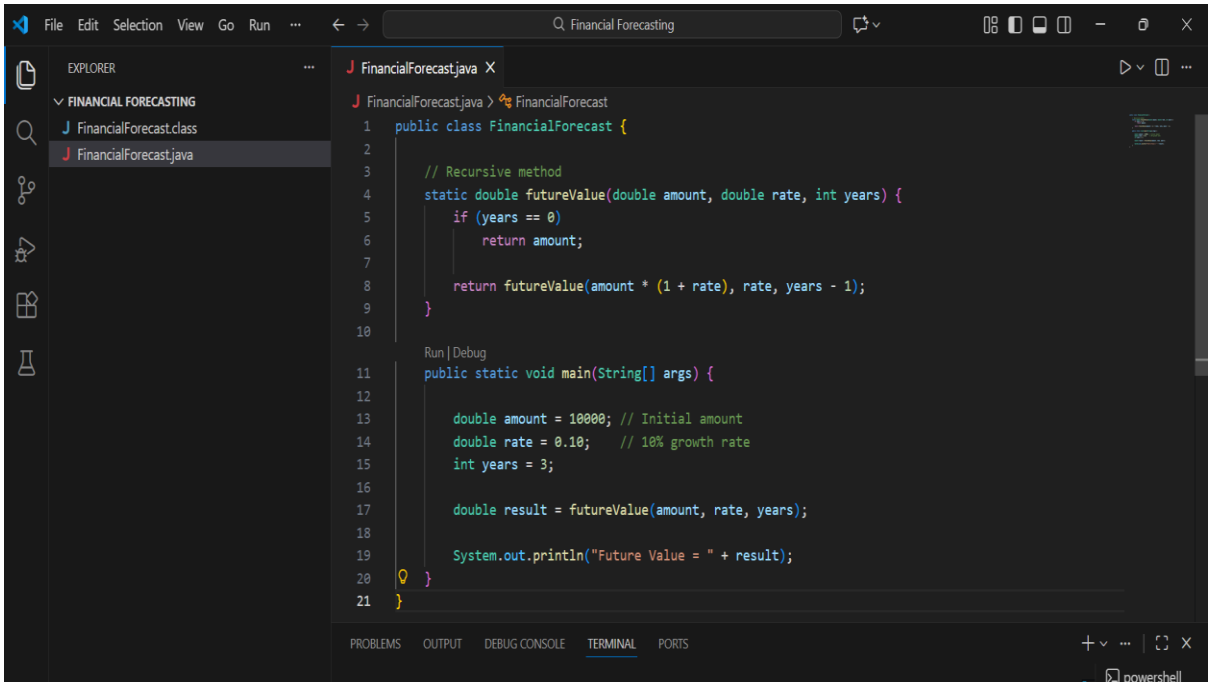


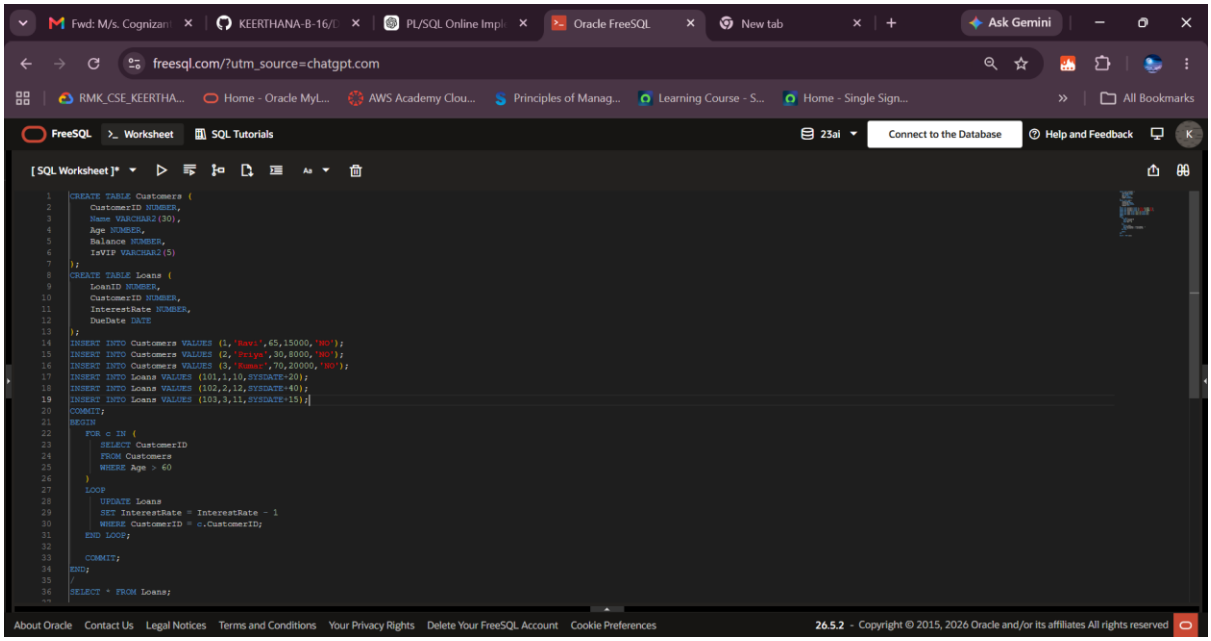
Exercise 7: Financial Forecasting

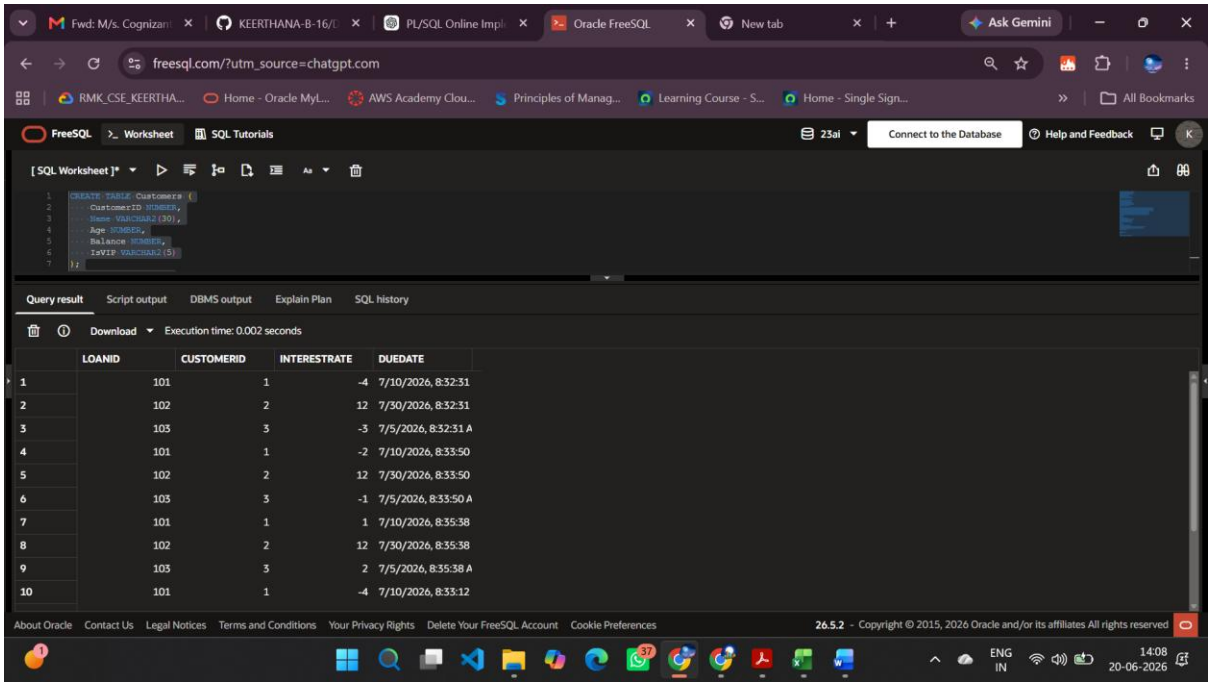


PL/SQL programming

Exercise 1: Control Structures

Scenario 1:




freesql.com/?utm_source=chatgpt.com
FreeSQL
Worksheet
SQL Tutorials
Connect to the Database
Help and Feedback
[SQL Worksheet]*
CREATE TABLE Customers (
CustomerID NUMBER,
Name VARCHAR2(30),
Age NUMBER,
Balance NUMBER,
IsVIP VARCHAR2(5)
);
Query result
Script output
DBMS output
Explain Plan
SQL history
Download
Execution time: 0.002 seconds
LOANID CUSTOMERID INTERESTRATE DUEDATE
1 101 1 -4 7/10/2026, 8:32:31
2 102 2 12 7/30/2026, 8:32:31
3 103 3 -3 7/5/2026, 8:32:31 A
4 101 1 -2 7/10/2026, 8:33:50
5 102 2 12 7/30/2026, 8:33:50
6 103 3 -1 7/5/2026, 8:33:50 A
7 101 1 1 7/10/2026, 8:35:38
8 102 2 12 7/30/2026, 8:35:38
9 103 3 2 7/5/2026, 8:35:38 A
10 101 1 -4 7/10/2026, 8:33:12
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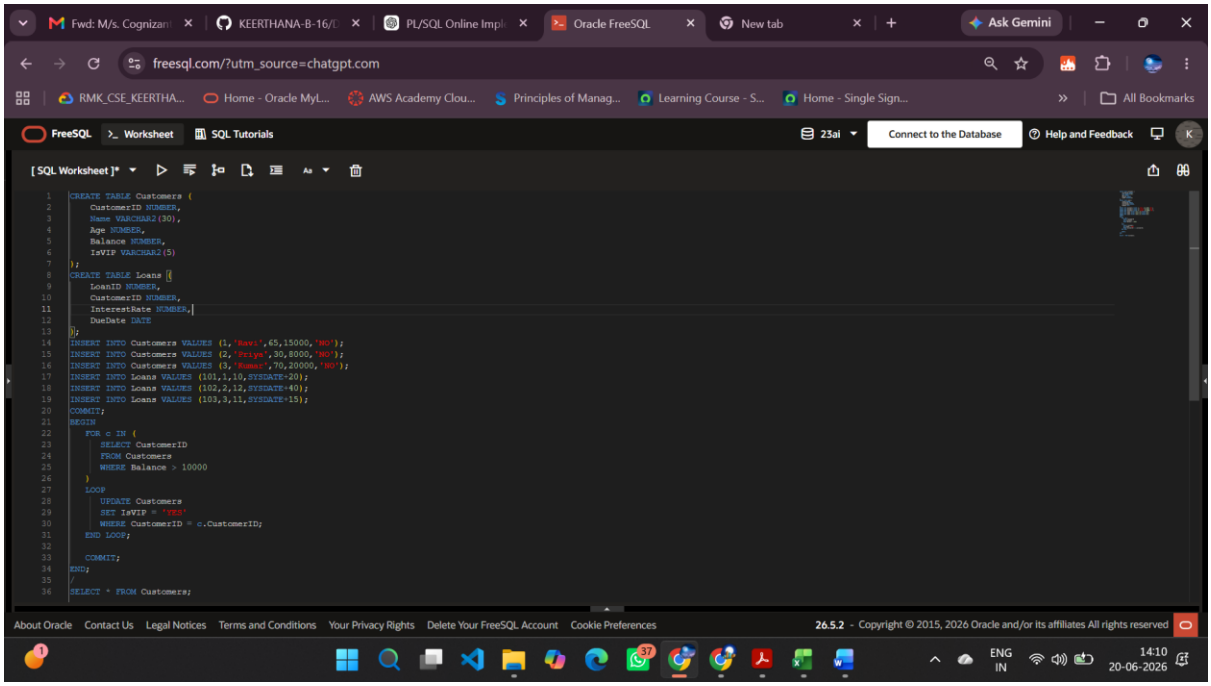
ENG
IN
14:08
20-06-2026

Scenario 2


freesql.com/?utm_source=chatgpt.com
FreeSQL
Worksheet
SQL Tutorials
Connect to the Database
Help and Feedback
[SQL Worksheet]*
CREATE TABLE Customers (
CustomerID NUMBER,
Name VARCHAR2(30),
Age NUMBER,
Balance NUMBER,
IsVIP VARCHAR2(5)
);
CREATE TABLE Loans (
LoanID NUMBER,
CustomerID NUMBER,
InterestRate NUMBER,
DueDate DATE
);
INSERT INTO Customers VALUES (1,'Ravi',65,15000,'NO');
INSERT INTO Customers VALUES (2,'Priya',30,8000,'NO');
INSERT INTO Customers VALUES (3,'Ramu',70,20000,'NO');
INSERT INTO Loans VALUES (101,1,10,SYSDATE-20);
INSERT INTO Loans VALUES (102,2,12,SYSDATE+40);
INSERT INTO Loans VALUES (103,3,11,SYSDATE+15);
COMMIT;
BEGIN
FOR c IN (
SELECT CustomerID
FROM Customers
WHERE Balance > 10000
)
LOOP
UPDATE Customers
SET IsVIP = 'YES'
WHERE CustomerID = c.CustomerID;
END LOOP;
COMMIT;
END;
/
SELECT * FROM Customers;
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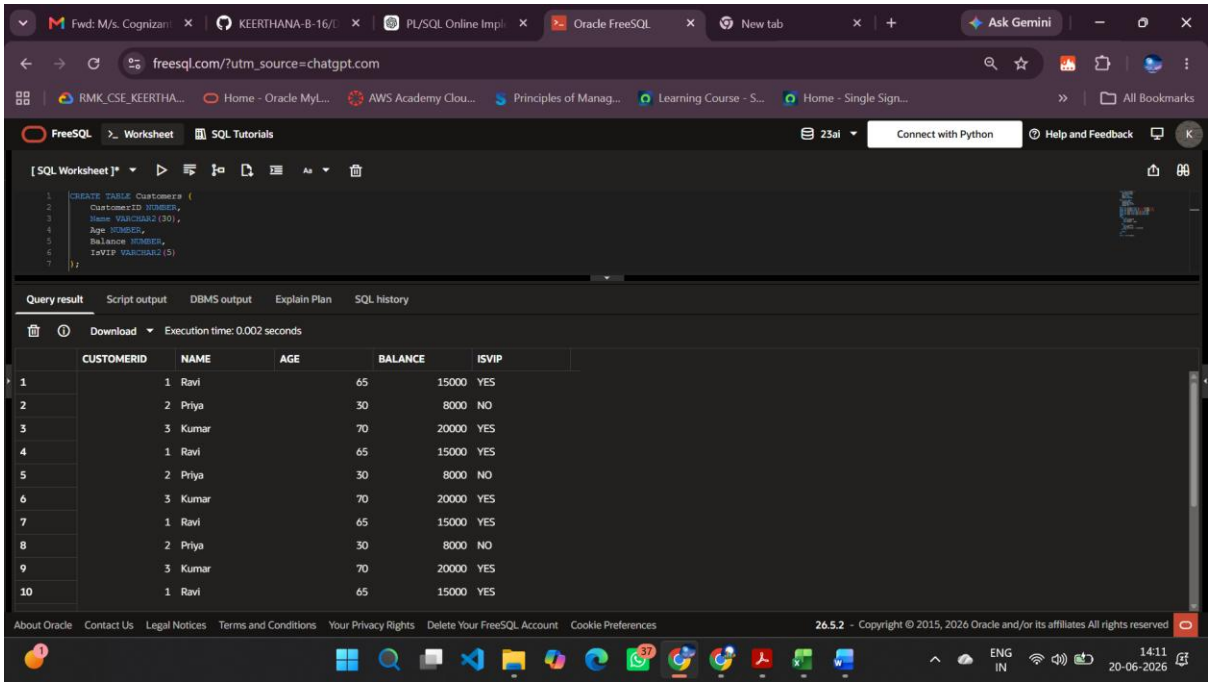
ENG
IN
14:10
20-06-2026


freesql.com/?utm_source=chatgpt.com
FreeSQL
Worksheet
SQL Tutorials
Connect with Python
Help and Feedback
[SQL Worksheet]*
CREATE TABLE Customers (
CustomerID NUMBER,
Name VARCHAR2(30),
Age NUMBER,
Balance NUMBER,
IsVIP VARCHAR2(5)
);
Query result
Script output
DBMS output
Explain Plan
SQL history
Download
Execution time: 0.002 seconds
CUSTOMERID NAME AGE BALANCE ISVIP
1 1 Ravi 65 15000 YES
2 2 Priya 30 8000 NO
3 3 Kumar 70 20000 YES
4 1 Ravi 65 15000 YES
5 2 Priya 30 8000 NO
6 3 Kumar 70 20000 YES
7 1 Ravi 65 15000 YES
8 2 Priya 30 8000 NO
9 3 Kumar 70 20000 YES
10 1 Ravi 65 15000 YES
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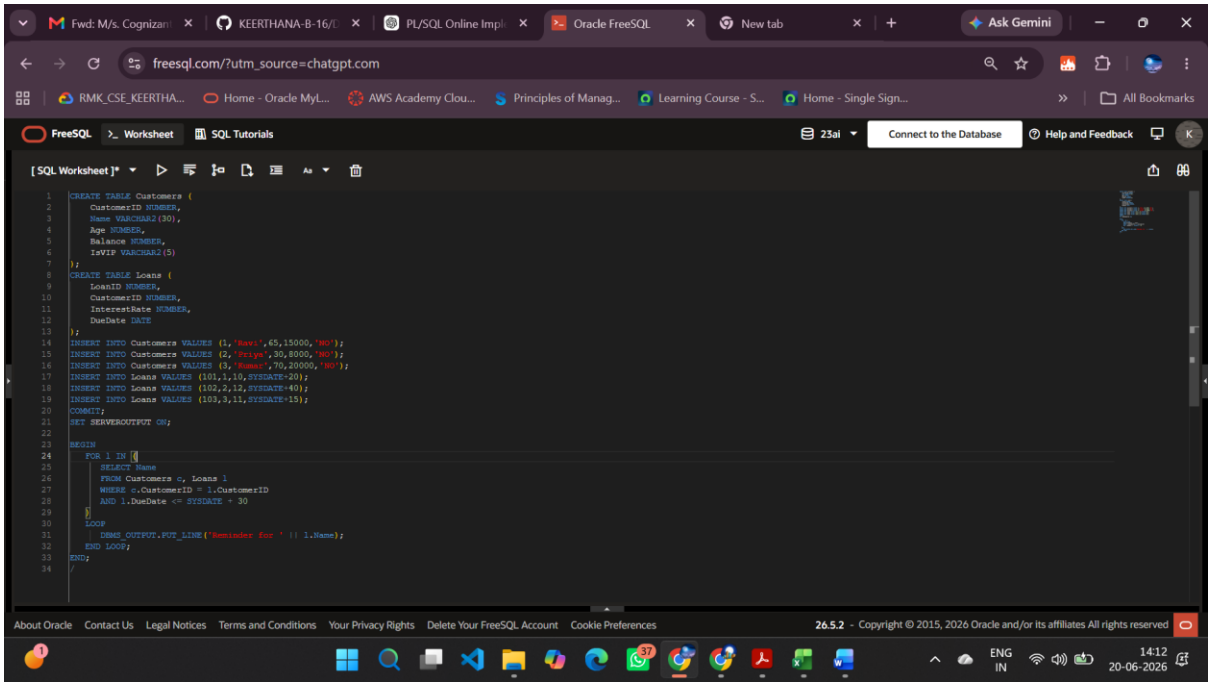
14:11
20-06-2026

Scenario 3


freesql.com/?utm_source=chatgpt.com
FreeSQL
Worksheet
SQL Tutorials
Connect to the Database
Help and Feedback
[SQL Worksheet]*
CREATE TABLE Customers (
CustomerID NUMBER,
Name VARCHAR2(30),
Age NUMBER,
Balance NUMBER,
IsVIP VARCHAR2(5)
);
CREATE TABLE Loans (
LoanID NUMBER,
CustomerID NUMBER,
InterestRate NUMBER,
DueDate DATE
);
INSERT INTO Customers VALUES (1,'Ravi',65,15000,'NO');
INSERT INTO Customers VALUES (2,'Priya',30,8000,'NO');
INSERT INTO Customers VALUES (3,'Kumar',70,20000,'NO');
INSERT INTO Loans VALUES (101,1,10,SYSDATE-20);
INSERT INTO Loans VALUES (102,2,12,SYSDATE+40);
INSERT INTO Loans VALUES (103,3,11,SYSDATE+15);
COMMIT;
SET SERVEROUTPUT ON;
BEGIN
FOR l IN (
SELECT Name
FROM Customers c, Loans l
WHERE c.CustomerID = l.CustomerID
AND l.DueDate <= SYSDATE + 30
)
LOOP
DBMS_OUTPUT.PUT_LINE('Reminder for ' || l.Name);
END LOOP;
END;
/
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ENG
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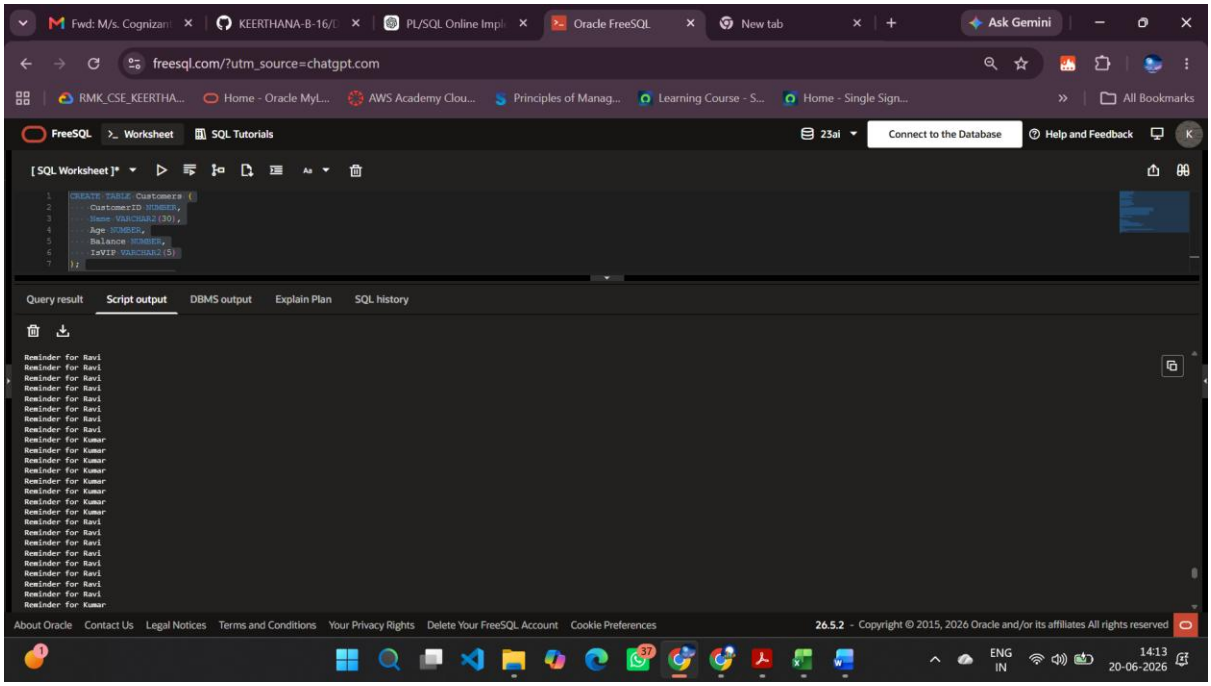
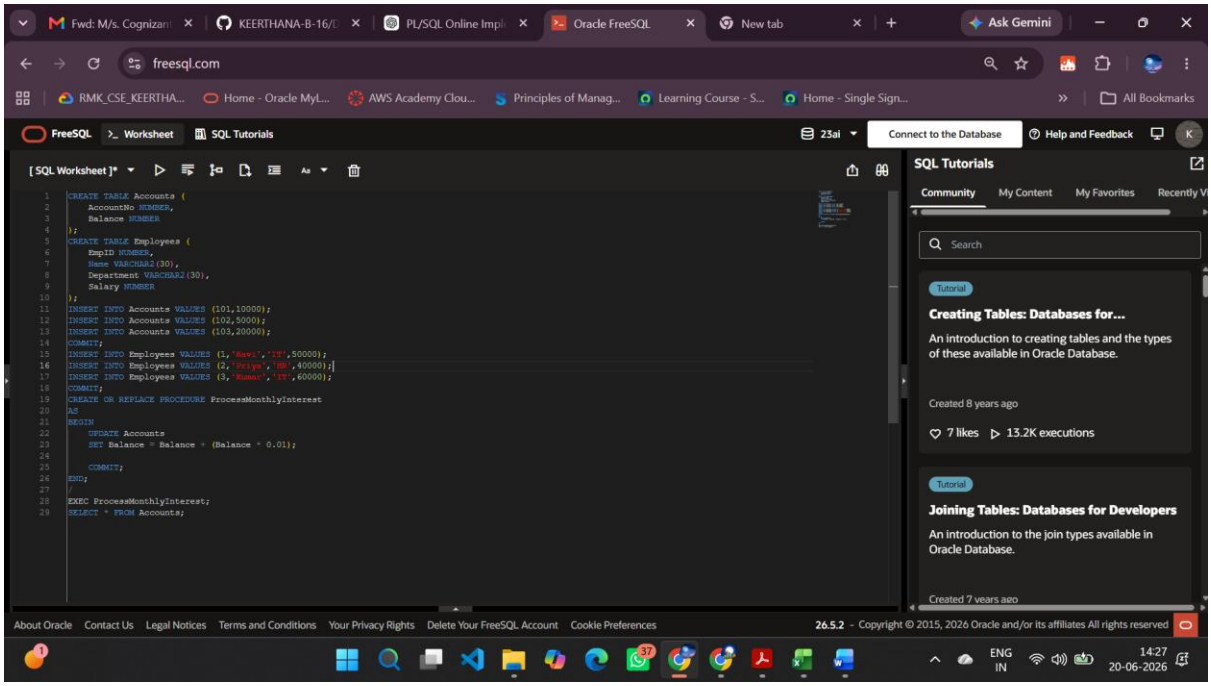
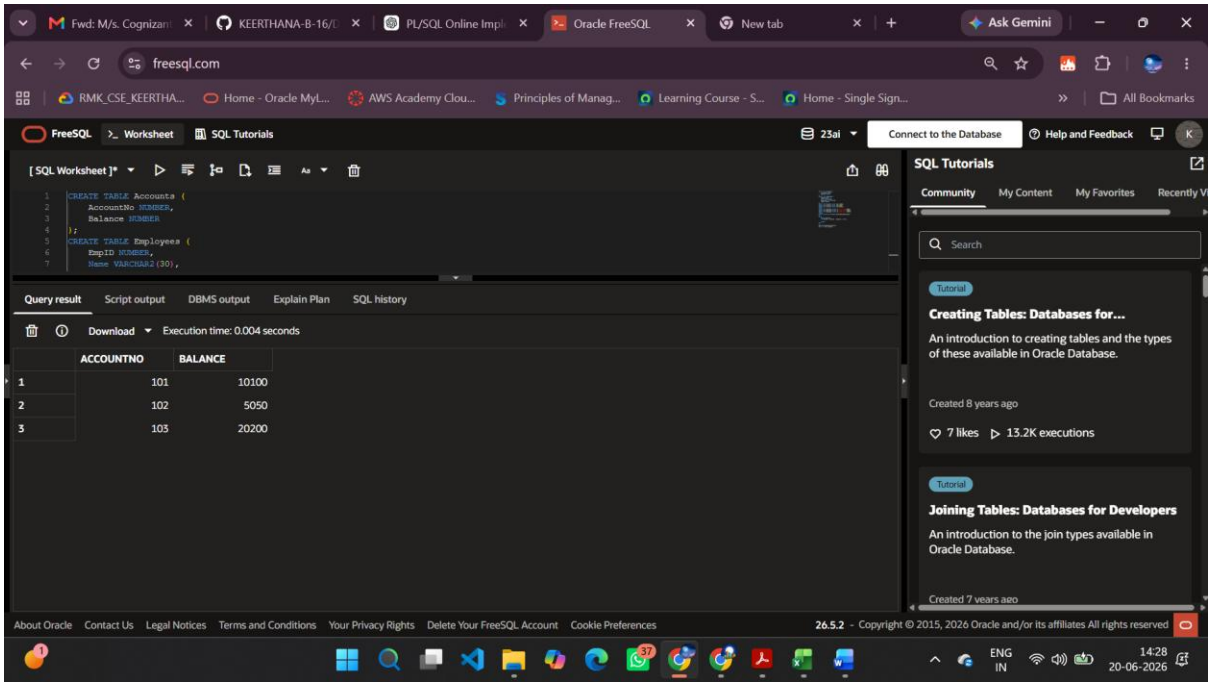
20-06-2026



Exercise 3: Stored Procedures

Scenario 1




freesql.com
FreeSQL
Worksheet
SQL Tutorials
Connect to the Database
Help and Feedback
[SQL Worksheet]*
CREATE TABLE Accounts (
AccountNo NUMBER,
Balance NUMBER
);
CREATE TABLE Employees (
EmpID NUMBER,
Name VARCHAR2(30),
Query result
Script output
DBMS output
Explain Plan
SQL history
Download
Execution time: 0.004 seconds
ACCOUNTNO
BALANCE
1 101 10100
2 102 5050
3 103 20200
SQL Tutorials
Community
My Content
My Favorites
Search
Tutorial
Creating Tables: Databases for...
An introduction to creating tables and the types of these available in Oracle Database.
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Tutorial
Joining Tables: Databases for Developers
An introduction to the join types available in Oracle Database.
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ENG
IN
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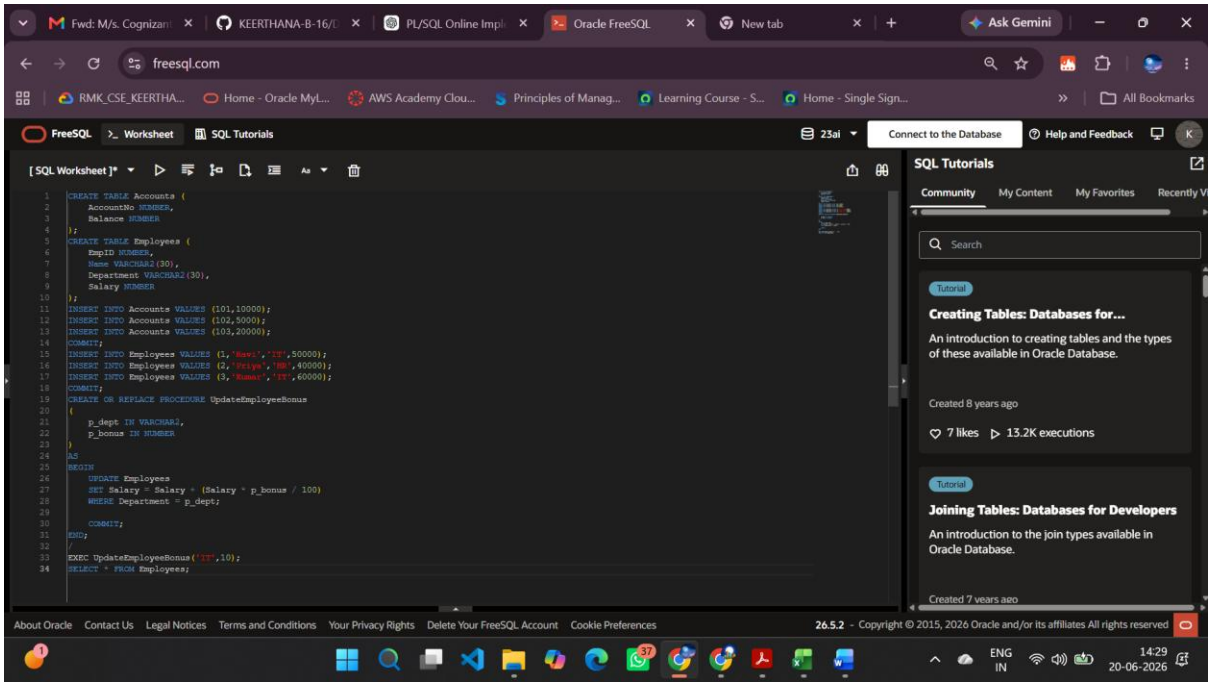
20-06-2026

Scenario 2:


freesql.com
FreeSQL
Worksheet
SQL Tutorials
Connect to the Database
Help and Feedback
[SQL Worksheet]*
CREATE TABLE Accounts (
AccountNo NUMBER,
Balance NUMBER
);
CREATE TABLE Employees (
EmpID NUMBER,
Name VARCHAR2(30),
Department VARCHAR2(30),
Salary NUMBER
);
INSERT INTO Accounts VALUES (101,10000);
INSERT INTO Accounts VALUES (102,5000);
INSERT INTO Accounts VALUES (103,20000);
COMMIT;
INSERT INTO Employees VALUES (1,'Ravi','IT',50000);
INSERT INTO Employees VALUES (2,'Priya','HR',40000);
INSERT INTO Employees VALUES (3,'Kumar','IT',60000);
COMMIT;
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus
(
p_dept IN VARCHAR2,
p_bonus IN NUMBER
)
AS
BEGIN
UPDATE Employees
SET Salary = Salary + (Salary * p_bonus / 100)
WHERE Department = p_dept;
COMMIT;
END;
/
EXEC UpdateEmployeeBonus('IT',10);
SELECT * FROM Employees;
SQL Tutorials
Community
My Content
My Favorites
Search
Tutorial
Creating Tables: Databases for...
An introduction to creating tables and the types of these available in Oracle Database.
Created 8 years ago
7 likes
13.2K executions
Tutorial
Joining Tables: Databases for Developers
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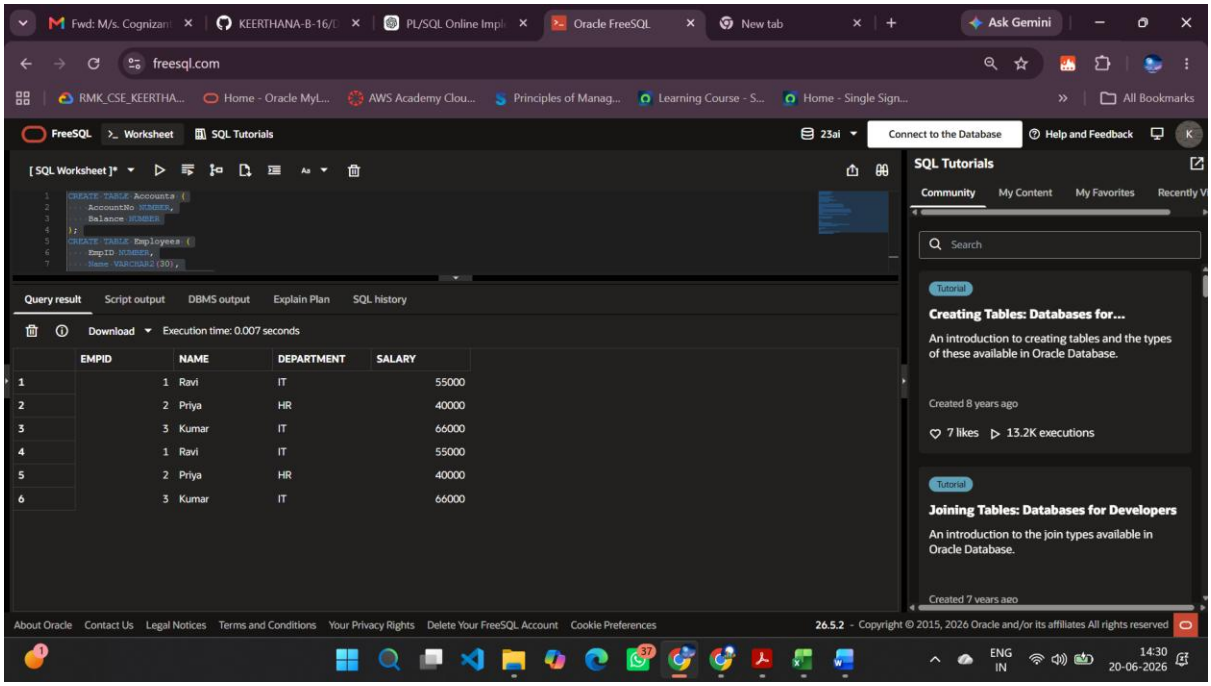
14:29
20-06-2026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[SQL Worksheet]*
CREATE TABLE Accounts (
AccountNo NUMBER,
Balance NUMBER
);
CREATE TABLE Employees (
EmpID NUMBER,
Name VARCHAR2(30),
Query result
Script output
DBMS output
Explain Plan
SQL history
Download
Execution time: 0.007 seconds
EMPID NAME DEPARTMENT SALARY
1 1 Ravi IT 55000
2 2 Priya HR 40000
3 3 Kumar IT 66000
4 1 Ravi IT 55000
5 2 Priya HR 40000
6 3 Kumar IT 66000
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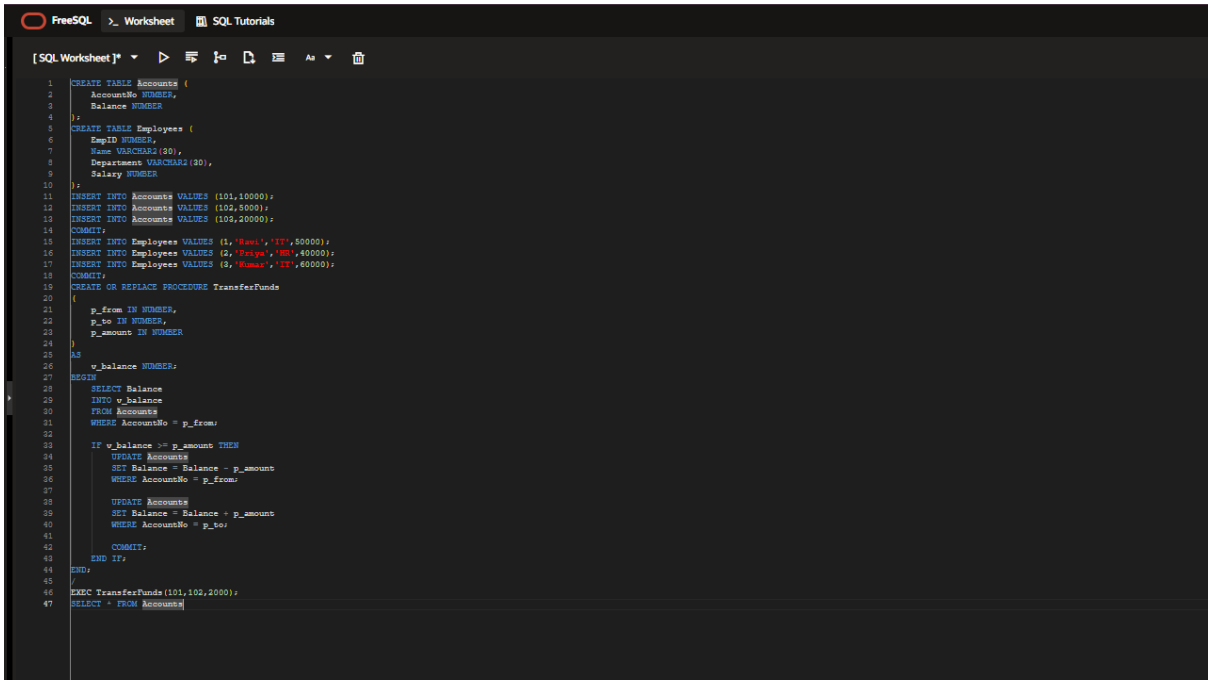
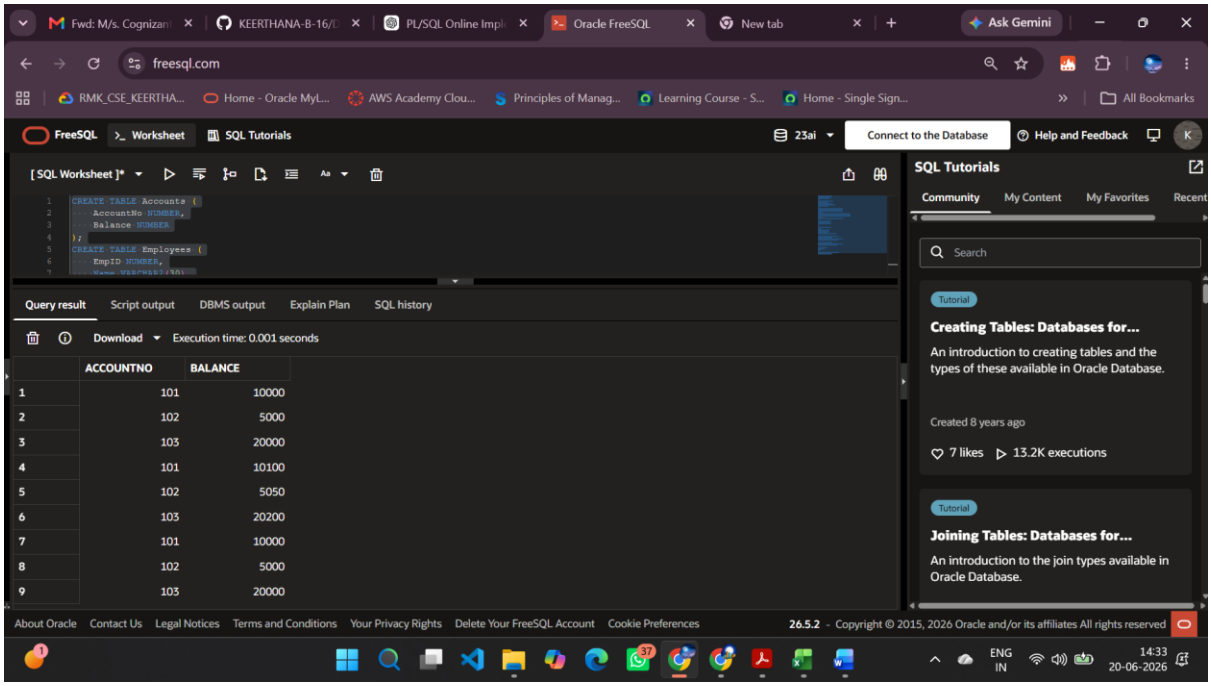
Scenario 3:


FreeSQL
Worksheet
SQL Tutorials
[SQL Worksheet]*
CREATE TABLE Accounts (
AccountNo NUMBER,
Balance NUMBER
);
CREATE TABLE Employees (
EmpID NUMBER,
Name VARCHAR2(30),
Department VARCHAR2(30),
Salary NUMBER
);
INSERT INTO Accounts VALUES (101,10000);
INSERT INTO Accounts VALUES (102,5000);
INSERT INTO Accounts VALUES (103,20000);
COMMIT;
INSERT INTO Employees VALUES (1,'Ravi','IT',50000);
INSERT INTO Employees VALUES (2,'Priya','HR',40000);
INSERT INTO Employees VALUES (3,'Kumar','IT',60000);
COMMIT;
CREATE OR REPLACE PROCEDURE TransferFunds
(
p_from IN NUMBER,
p_to IN NUMBER,
p_amount IN NUMBER
)
AS
v_balance NUMBER;
BEGIN
SELECT Balance
INTO v_balance
FROM Accounts
WHERE AccountNo = p_from;
IF v_balance >= p_amount THEN
UPDATE Accounts
SET Balance = Balance - p_amount
WHERE AccountNo = p_from;
UPDATE Accounts
SET Balance = Balance + p_amount
WHERE AccountNo = p_to;
COMMIT;
END IF;
END;
/
EXEC TransferFunds(101,102,2000);
SELECT * FROM Accounts



JUnit Testing Exercises

Exercise 1: Setting Up JUnit
pom.xml

```
<dependency>
    <groupId>junit</groupId>
    <artifactId>junit</artifactId>
    <version>4.13.2</version>
    <scope>test</scope>
</dependency>
```

CalculatorTest.java

```
import static org.junit.Assert.assertEquals;
import org.junit.Test;
public class CalculatorTest {
    @Test
    public void testAdd() {
        Calculator c = new Calculator();
        assertEquals(5, c.add(2, 3));
    }
}
```

OUTPUT:

Tests run: 1

Failures: 0

Errors: 0

BUILD SUCCESS

Exercise 3: Assertions in JUnit
AssertionsTest.java

```
import static org.junit.Assert.*;
import org.junit.Test;
public class AssertionsTest {
    @Test
    public void testAssertions() {
        // Assert equals
        assertEquals(5, 2 + 3);
        // Assert true
        assertTrue(5 > 3);
        // Assert false
        assertFalse(5 < 3);
        // Assert null
        assertNull(null);
        // Assert not null
        assertNotNull(new Object());
    }
}
```

OUTPUT

Tests run: 1

Failures: 0

Errors: 0

BUILD SUCCESS

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in Junit

Calculator.java

```
public class Calculator {
    public int add(int a, int b) {
        return a + b;
    }
}
```

CalculatorTest.java

```
import static org.junit.Assert.assertEquals;
import org.junit.After;
import org.junit.Before;
import org.junit.Test;
public class CalculatorTest {
    private Calculator calculator;
    @Before
    public void setUp() {
        calculator = new Calculator();
        System.out.println("Setup completed");
    }
    @Test
    public void testAdd() {
        // Arrange
        int a = 2;
        int b = 3;
        // Act
        int result = calculator.add(a, b);
        // Assert
        assertEquals(5, result);
    }
    @After
    public void tearDown() {
```

```
        calculator = null;
        System.out.println("Teardown completed");
    }
}
```

OUTPUT

Setup completed

Teardown completed

Tests run: 1

Failures: 0

Errors: 0

BUILD SUCCESS

Mockito exercises

Exercise 1: Mocking and Stubbing

ExternalApi.java

```
public interface ExternalApi {
    String getData();
}
```

MyService.java

```
public class MyService {
    private ExternalApi api;
    public MyService(ExternalApi api) {
        this.api = api;
    }
    public String fetchData() {
        return api.getData();
    }
}
```

MyServiceTest.java

```
import static org.junit.Assert.assertEquals;
import static org.mockito.Mockito.*;
import org.junit.Test;
import org.mockito.Mockito;
public class MyServiceTest {
    @Test
    public void testExternalApi() {
        ExternalApi mockApi = Mockito.mock(ExternalApi.class);
        when(mockApi.getData()).thenReturn("Mock Data");
        MyService service = new MyService(mockApi);
        String result = service.fetchData();
        assertEquals("Mock Data", result);
    }
```

```
}
```
OUTPUT
```
Tests run: 1
Failures: 0
Errors: 0
BUILD SUCCESS
```

Exercise 2: Verifying Interactions

ExternalApi.java
```
public interface ExternalApi {
   String getData();
}
```
MyService.java
```
public class MyService {
   private ExternalApi api;
   public MyService(ExternalApi api) {
      this.api = api;
   }
   public String fetchData() {
      return api.getData();
   }
}
```
MyServiceTest.java
```
import static org.mockito.Mockito.*;
import org.junit.Test;
import org.mockito.Mockito;
public class MyServiceTest {
   @Test
   public void testVerifyInteraction() {
      // Create mock object
      ExternalApi mockApi = Mockito.mock(ExternalApi.class);
      // Create service object
      MyService service = new MyService(mockApi);
      // Call method
      service.fetchData();
      // Verify interaction
      verify(mockApi).getData();
   }
}
```
OUTPUT
```
Tests run: 1
Failures: 0
Errors: 0
BUILD SUCCESS
```

SLF4J logging framework

Exercise 1: Logging Error Messages and Warning Levels

pom.xml

```
<dependency>
   <groupId>org.slf4j</groupId>
   <artifactId>slf4j-api</artifactId>
   <version>1.7.30</version>
</dependency>

<dependency>
   <groupId>ch.qos.logback</groupId>
   <artifactId>logback-classic</artifactId>
   <version>1.2.3</version>
</dependency>
```

LoggingExample.java

```
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
public class LoggingExample {
   private static final Logger logger =
         LoggerFactory.getLogger(LoggingExample.class);
   public static void main(String[] args) {
      logger.error("This is an error message");
      logger.warn("This is a warning message");
   }
}
```

OUTPUT

```
ERROR LoggingExample - This is an error message
WARN  LoggingExample - This is a warning message
```